

Aircraft Safety Analysis

Data-Driven Recommendations for Fleet Procurement

Summary

This project analyzes aviation crash data from 2018-2022 to identify the safest aircraft types for fleet expansion. By examining over 2,000 accidents across 100+ aircraft models, I uncovered critical safety patterns that inform procurement decisions.

Key Findings:

- Boeing and Cessna lead in total accidents, with Cessna showing higher fatality rates
- Industry-wide accidents declined 26% from 2018 to 2021
- Airbus demonstrates exceptional safety with almost entirely non-fatal outcomes
- ATR, Embraer, and Bombardier show significantly lower accident counts

Recommendation:

Prioritize Airbus, Embraer, and ATR models for new acquisitions to minimize operational risk and enhance passenger safety.

Outline



01 Business Problem



02 Data



03 Methods



04 Results



05 Conclusions

01. Business Problem

How can we identify the safest aircraft types to minimize operational risk and maximize passenger safety for our expanding aviation fleet?

Key Stakeholders:

- Company Shareholders - Investment decisions and risk assessment
- Procurement Managers - Aircraft selection and purchasing
- Operations and Safety Teams - Fleet management and safety protocols

Business Objectives:

- Reduce accident-related costs and liability exposure
- Enhance brand reputation as a safety-conscious operator
- Make data-driven decisions for long-term fleet diversification

02. Data

Data Source:

Aviation Crashed Flights Data from Kaggle covering January 2018 to January 2022

Key Data Fields:

- Accident Date (acc.date) - Date of incident
- Type - Aircraft model/type
- Registration (reg) - Unique aircraft identification
- Operator - Company operating the aircraft
- Fatalities (fat) - Number of fatalities
- Location - Geographic location of crash
- Damages (dmg) - Severity (Substantial, Write-off, No Damage)

03. Methods

Data Preparation

- • Handled missing values
- • Standardized aircraft type names
- • Converted date fields to datetime
- • Categorized damage severity

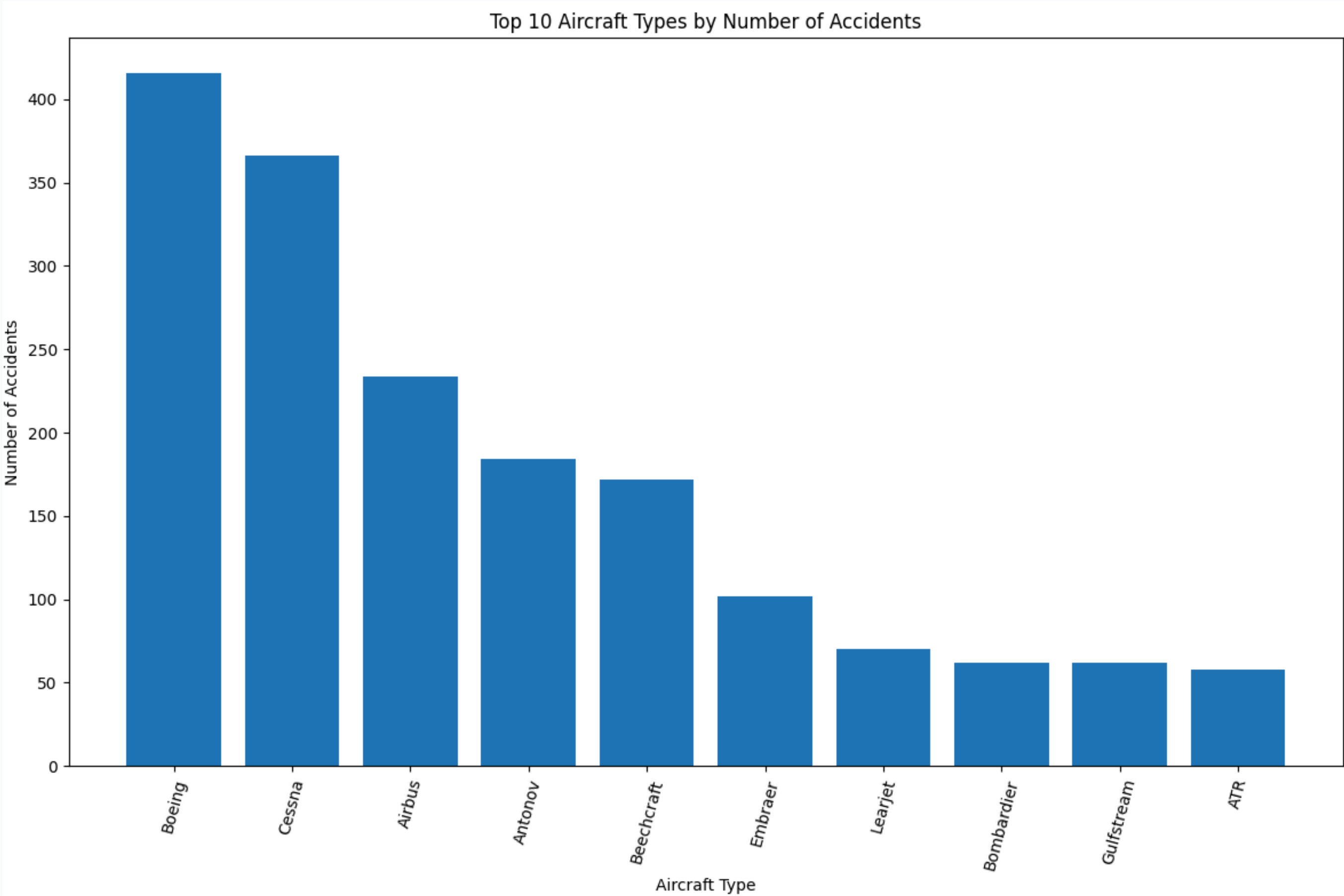
Analysis Methods

- • Frequency analysis by aircraft type
- • Time-series trend analysis
- • Fatality distribution analysis
- • Fatal vs non-fatal comparison

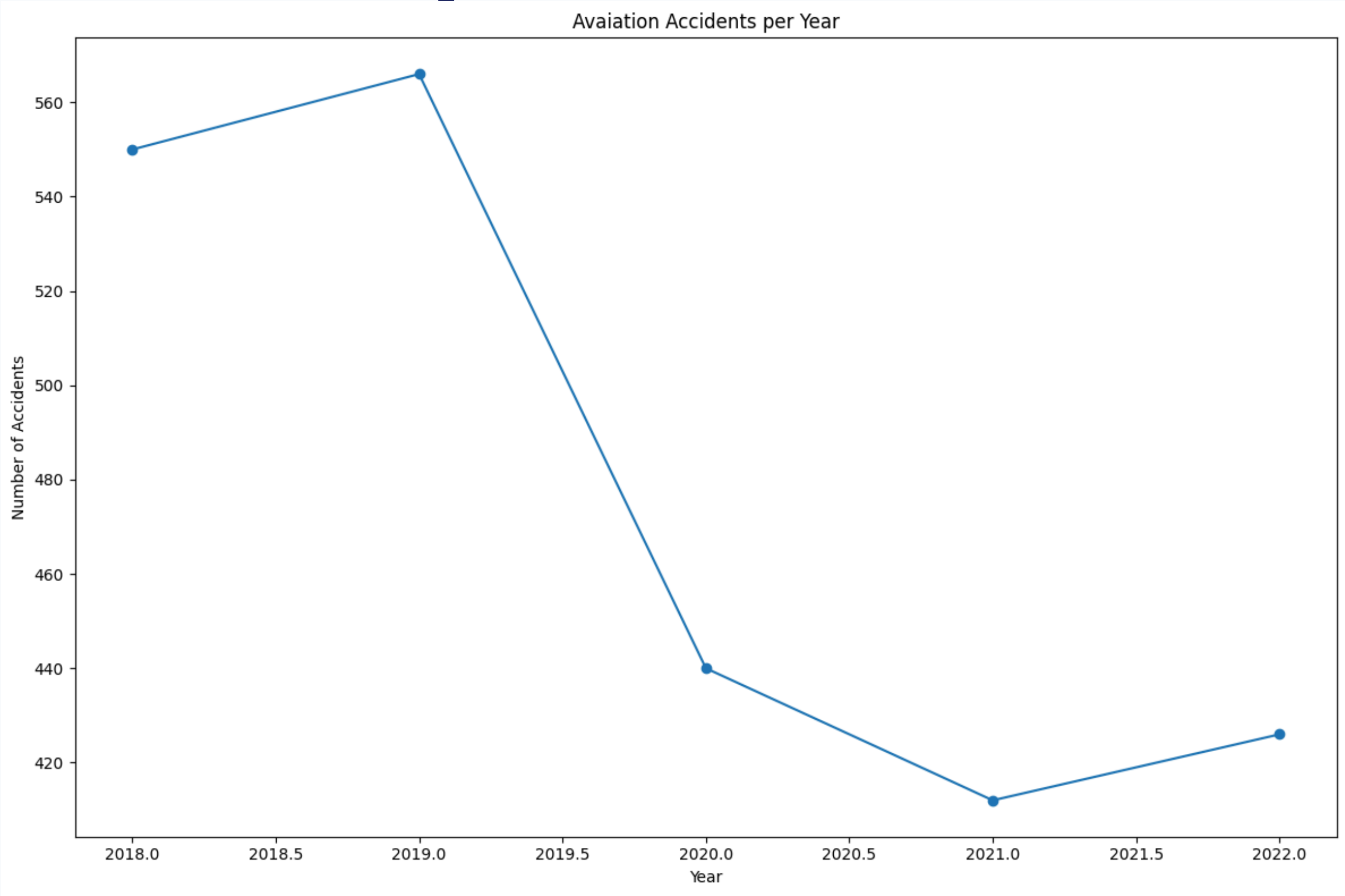
Visualization Techniques

- • Bar charts for accident frequency comparison across aircraft types
- • Line charts for temporal trend analysis (2018-2022)
- • Box plots for fatality distribution and outlier detection
- • Stacked bar charts for fatal vs non-fatal accident breakdown

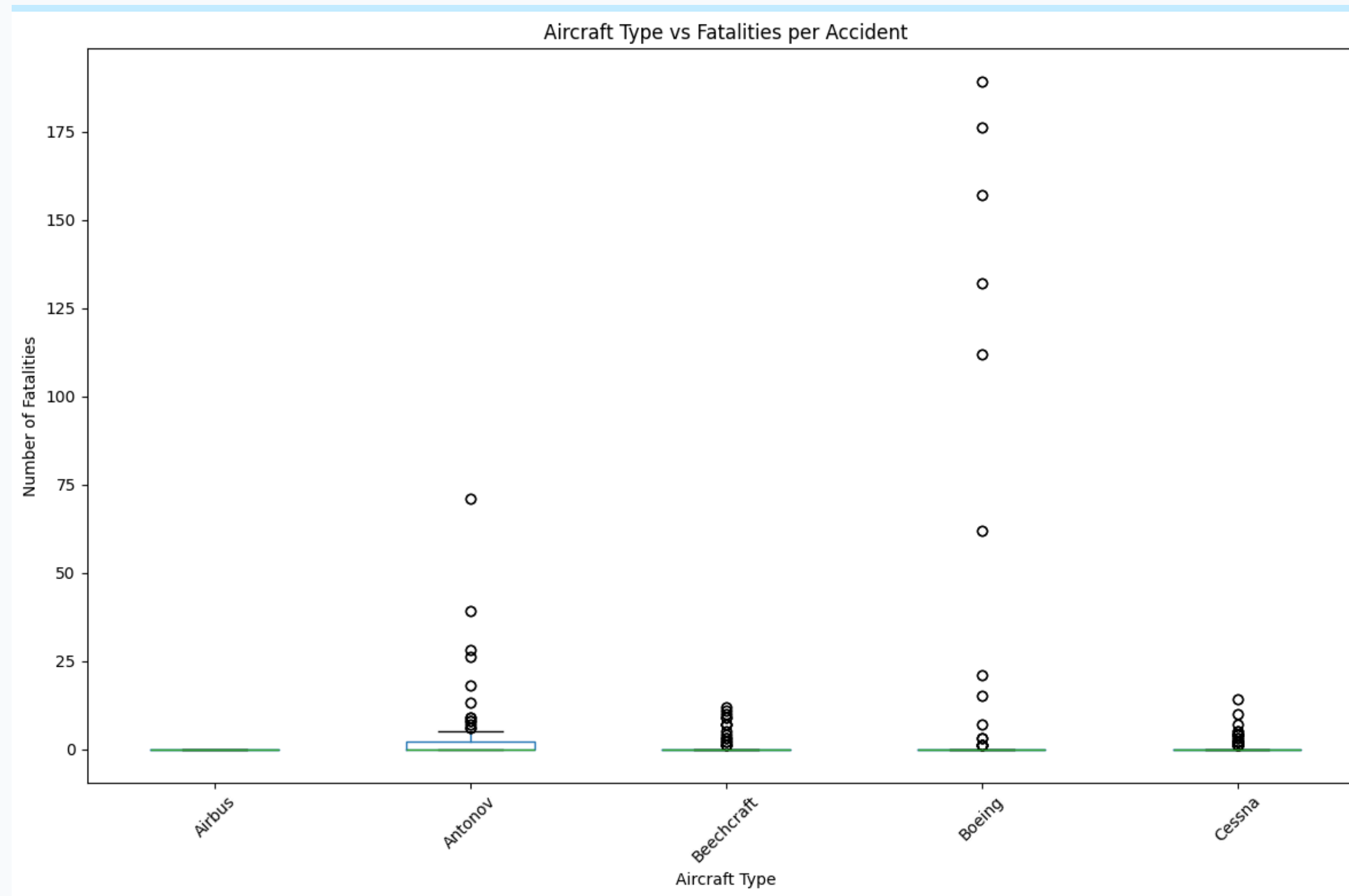
Results: Aircraft Types by Accidents



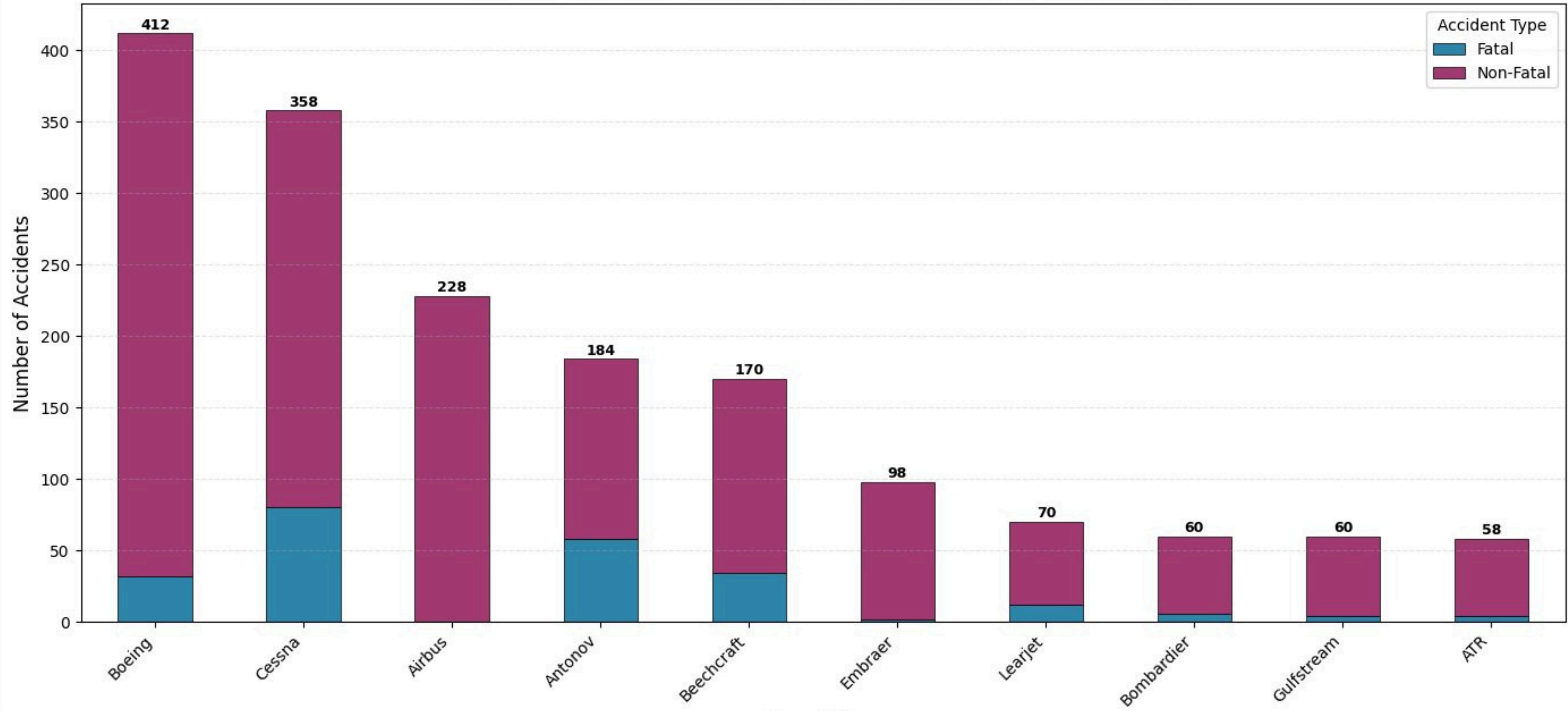
Results: Safety Trends Over Time



Results: Fatality Distribution



Results: Fatal vs Non-Fatal Breakdown



04. Results: How Well Do They Solve the Problem?

✓ Clear Risk Identification

Analysis successfully identified high-risk aircraft (Boeing, Cessna) and safer alternatives (Airbus, Embraer, ATR), directly addressing the core business question.

✓ Actionable Insights

Results provide concrete recommendations for procurement decisions with clear evidence of safety performance differences between manufacturers.

✓ Quantifiable Business Value

26% reduction in accidents industry-wide demonstrates measurable safety improvements. Selecting top-performing manufacturers can further reduce risk and associated costs.

05. Conclusions

Business Recommendations:

- Prioritize Airbus, Embraer, and ATR for new aircraft acquisitions
- Implement enhanced safety protocols for existing Boeing fleet
- Phase out older Cessna models with higher fatality rates
- Leverage safety record as competitive advantage in marketing

Project Limitations:

- Analysis limited to 2018-2022 data; newer models may not have sufficient history
- Does not account for exposure (flight hours, number of aircraft in operation)
- Root cause analysis of accidents not included in current scope

Future Improvement Ideas:

- Incorporate flight hours data for accident rate per operation
- Analyze accident causes by aircraft type for deeper insights
- Regional safety analysis to optimize route planning

Thank You!

Questions?

Email: moonwaangela@gmail.com

GitHub: @MoonwaMasaki

LinkedIn: <https://www.linkedin.com/in/angela-masaki-396043215/>